

TOPIC 4: NETWORK TOPOLOGIES

1. Meaning of Topology

Network topology is the way in which computers and devices are arranged and connected in a network – the layout or structure of the network.

2. Common Types

- **Bus topology** – all devices share a single main cable. Simple and cheap for small networks; if the main cable fails, the whole network can fail.
- **Star topology** – all devices connect to a central switch or hub. Easy to add devices; failure of one cable does not stop the others; if the centre fails, the network fails. Very common in schools and offices.
- **Ring topology** – devices connected in a closed loop. Data travels in one direction; a break in the ring can disrupt the network unless there is a backup path.

3. Advantages and Disadvantages (Brief)

Star is preferred for most LANs because it is reliable and easy to manage. Bus is simple but less robust. Ring is less common in modern small networks.

4. Summary

Topology describes how devices are connected. Bus, star and ring each have strengths and weaknesses; star is the most widely used in practice.

End of Topic 4